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★ REVIEWS

Micronutrients for Attention-Deficit/Hyperactivity Disorder in Youth: A Placebo-Controlled Randomized Clinical Trial

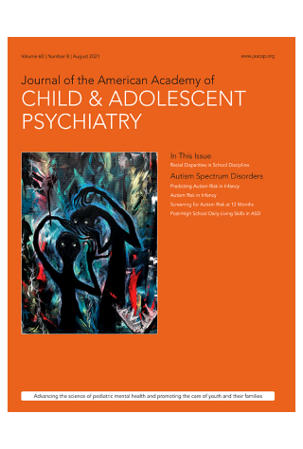
Abstract

Objective: To evaluate whether micronutrients (vitamins/minerals) benefit attention-deficit/hyperactivity disorder (ADHD) and irritability in a North American pediatric sample.

Method: A three-site 8-week placebo-controlled randomized clinical trial of micronutrients was conducted in unmedicated children ages 6-12 with ADHD and at least one impairing irritability symptom by parent report on the Child and Adolescent Symptom Inventory-5 (CASI-5). A priori-defined primary outcomes were Clinical Global Impression-Improvement (CGI-I) (CGI-I of 1 or 2 = treatment responder) and parent-rated CASI-5 composite score of ADHD, oppositional defiant, disruptive mood dysregulation, and peer conflict symptoms, including impairment scores.

Results: Of 135 randomized (mean age 9.8 years), 126 (93%) comprised the modified intention-to-treat population. Blinding was maintained. For the CGI-I, 54% of the micronutrient and 18% of the placebo group were responders (Risk Ratio=2.97, 97.5% CI: 1.50, 5.90, p<0.001). CASI-5 composite scores improved significantly for both groups (p<0.01) with a mean change of -0.31 (95% CI: -0.39, -0.23) in the micronutrient group and a mean change of -0.28 (95% CI: -0.38, -0.19) in the placebo group. But the between group difference was not significant (mean change = -0.02; 97.5% CI: -0.16, 0.12, ES = 0.07, p=0.70). The micronutrient group grew six millimeters more than the placebo group (p=0.002). No serious adverse events nor clinically significant changes from baseline in blood and urine tests occurred.

Conclusion: Micronutrients showed global benefit over placebo by blinded clinician rating, but not by parent-report CASI-5 composite rating in a population with ADHD and irritability. Micronutrients showed greater height growth. Micronutrients were well tolerated and the majority adhered to the number of capsules prescribed. This RCT replicates safety and efficacy reported for ADHD in two smaller trials of a similar formula containing all vitamins and known essential minerals in amounts between the Recommended Dietary Allowance and Upper Tolerable Intake Level.



Micronutrients for Attention-Deficit/Hyperactivity Disorder in Youth: A Placebo-Controlled Randomized Clinical Trial

Johnstone JM, Hatsu I, Tost G, Srikanth P, Eiterman LP, Bruton A, Ast HK, Robinette LM, Stern MM, Millington EG, Gracious B, Hughes AJ, Leung BM, Arnold LE. Micronutrients for Attention-Deficit/Hyperactivity Disorder in Youth: A Placebo-Controlled Randomized Clinical Trial. J Am Acad Child Adolesc Psychiatry. 2021 Jul 21:S0890-8567(21)00473-1.

FULL TEXT

Clinician guidelines for the treatment of psychiatric disorders with nutraceuticals and phytoceuticals: The World Federation of Societies of Biological Psychiatry (WFSBP) and Canadian Network for Mood and Anxiety Treatments (CANMAT) Taskforce

The 'Micronutrient Formula' rated by this international taskforce was Daily Essential Nutrients, and was given an 'Evidence Grade' of 'A', due to a statistically significant meta-analysis (*k* = 2, *n* = 173) with statistically significant RCTs in adult and child samples. They gave the following supporting statements for Daily Essential Nutrients:

Meta-analytic level results have shown supportive evidence for efficacy in ADHD as a monotherapy

More replicated evidence required (in both adults and children)

This particular micronutrient formula's efficacy cannot necessarily be extended to other multi-nutrient formulas

Dosing may need to be supervised (and titrated) via a health profession

Cost and compliance may be an issue due to a recommended dosage of 8–12 capsules per day

Acceptable safety data

Abstract

Objectives: The therapeutic use of nutrient-based 'nutraceuticals' and plant-based 'phytoceuticals' for the treatment of mental disorders is common; however, despite recent research progress, there have not been any updated global clinical guidelines since 2015. To address this, the World Federation of Societies of Biological Psychiatry (WFSBP) and the Canadian Network for Mood and Anxiety Disorders (CANMAT) convened an international taskforce involving 31 leading academics and clinicians from 15 countries, between 2019 and 2021. These guidelines are aimed at providing a definitive evidence-informed approach to assist clinicians in making decisions around the use of such agents for major psychiatric disorders. We also provide detail on safety and tolerability, and clinical advice regarding prescription (e.g. indications, dosage), in addition to consideration for use in specialised populations.

Methods: The methodology was based on the WFSBP guidelines development process. Evidence was assessed based on the WFSBP grading of evidence (and was modified to focus on Grade A level evidence - meta-analysis or two or more RCTs - due to the breadth of data available across all nutraceuticals and phytoceuticals across major psychiatric disorders). The taskforce assessed both the 'level of evidence' (LoE) (i.e. meta-analyses or RCTs) and the assessment of the direction of the evidence, to determine whether the intervention was 'Recommended' (+++), 'Provisionally Recommended' (++), 'Weakly Recommended' (+), 'Not Currently Recommended' (+/-), or 'Not Recommended' (-) for a particular condition. Due to the number of clinical trials now available in the field, we firstly examined the data from our two meta-reviews of meta-analyses (nutraceuticals conducted in 2019, and phytoceuticals in 2020). We then performed a search of additional relevant RCTs and reported on both these data as the primary drivers supporting our clinical recommendations. Lower levels of evidence, including isolated RCTs, open label studies, case studies, preclinical research, and interventions with only traditional or anecdotal use, were not assessed.

Conclusions: Based on the current data and clinician input, a range of nutraceuticals and phytoceuticals were given either a supportive recommendation or a provisional recommendation across a range of various psychiatric disorders. However several had only a weak endorsement for potential use; for a few it was not possible to reach a clear recommendation direction, largely due to mixed study findings; while some other agents showed no obvious therapeutic benefit and were clearly not recommended for use. It is the intention of these guidelines to inform psychiatric/medical, and health professional practice globally.

Clinician guidelines for the treatment of psychiatric disorders with nutraceuticals and phytoceuticals: The World Federation of Societies of Biological Psychiatry (WFSBP) and Canadian Network for Mood and Anxiety Treatments (CANMAT) Taskforce

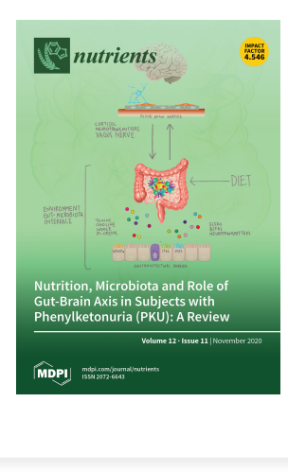
Sarris J, Ravindran A, Yatham LN, Marx W, Rucklidge JJ, McIntyre RS, Akhondzadeh S, Benedetti F, Caneo C, Cramer H, Cribb L, de Manincor M, Dean O, Deslandes AC, Freeman MP, Gangadhar B, Harvey BH, Kasper S, Lake J, Lopresti A, Lu L, Metri NJ, Mischoulon D, Ng CH, Nishi D, Rahimi R, Seedat S, Sinclair J, Su KP, Zhang ZJ, Berk M. Clinician guidelines for the treatment of psychiatric disorders with nutraceuticals and phytoceuticals: The World Federation of Societies of Biological Psychiatry (WFSBP) and Canadian Network for Mood and Anxiety Treatments (CANMAT) Taskforce. World J Biol Psychiatry. 2022 Jul;23(6):424-455.

FULL TEXT

Multinutrients for the Treatment of Psychiatric Symptoms in Clinical Samples: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

Abstract

This systematic review and meta-analysis focused on randomized controlled trials (RCT) of multinutrients consisting of at least four vitamins and/or minerals as interventions for participants with psychiatric symptoms. A systematic search identified 16 RCTs that fit the inclusion criteria (n = 1719 participants) in six psychiatric categories: depression, post-disaster stress, antisocial behavior, behavioral deficits in dementia, attention-deficit/hyperactivity disorder, and autism. Grading of Recommendations, Assessment, Development and Evaluations (GRADE) was used to rate the evidence base. Significant clinical benefit was assessed using minimal clinically important differences (MIDs). Due to heterogeneity in participants, multinutrient formulas, outcome measures, and absence of complete data, only the Attention-Deficit/Hyperactivity Disorder (ADHD) category was eligible for meta-analyses. In ADHD populations, statistically and clinically significant improvements were found in global functioning, Mean Difference (MD) -3.3, p = 0.001, MID -3.26; Standardized Mean Difference (SMD) -0.49 p = 0.001 MD -0.5), clinician ratings of global improvement (MD -0.58, p = 0.001, MID -0.5) and ADHD improvement (MD -0.54, p = 0.002, MID -0.5), and clinician (but not observer) measures of ADHD inattentive symptoms (MD -1.53, p = 0.05, MID -0.5). Narrative synthesis also revealed a pattern of benefit for global measures of improvement, for example: in autism, and in participants with behavioral deficits in dementia. Post-natural disaster anxiety and the number of violent incidents in prison populations also improved. Broad-spectrum formulas (vitamins + minerals) demonstrated more robust effects than formulas with fewer ingredients. This review highlights the need for robust methodology-RCTs that report full data, including means and standard deviations for all outcomes-in order to further elucidate the effects of multinutrients for psychiatric symptoms.



Multinutrients for the Treatment of Psychiatric Symptoms in Clinical Samples: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

Jeanette M Johnstone, Andrew Hughes, Joshua Z Goldenberg, Amy R Romijn, Julia J Rucklidge. Nutrients. 2020 Nov 4;12(11):3394.

FULL TEXT

Safety and tolerability of Micronutrient formulations.

Scientists compiled safety data from all published and unpublished studies that used a micronutrient formulation co-developed† by Hardy Nutritionals® founder David Hardy which has been widely researched in the mental health field. Biological safety data from six sources, including 144 children and adults, showed no occurrences of clinically meaningful negative outcomes/effects or abnormal blood tests that could be attributed to toxicity. Adverse event information from six studies involving 157 children and adults included reports of minor, transitory headache and nausea. Only one of the studies permitted a direct comparison between tolerability of the micronutrient treatment and medications: children and adults treated with micronutrients showed significantly fewer adverse events and significantly less weight gain compared with those taking medications.

Systematic review of safety and tolerability of
a complex micronutrient formula used in mental health.

Simpson JS, Crawford SG, Goldstein ET, Field C, Burgess E, Kaplan BJ. BioMed Central Psychiatry. 2011 Apr 18;11:62.

FULL TEXT

Micronutrient formulation outperforms medications in 88 patients with autism.

Researchers compared psychiatric medication and micronutrient treatment approaches in 88 children with autism spectrum disorder. The micronutrient group (taking predominately a micronutrient formulation co-developed† by Hardy Nutritionals® founder David Hardy) improved significantly more than the medication group in key ways, including notably reduced irritability, anger, and intensity of self-injurious behaviors. The micronutrient-treated group also had dramatically lower treatment side-effects.

Micronutrients versus standard medication management in autism: a naturalistic case-control study.

Mehl-Madrona L, Leung B, Kennedy C, Paul S, Kaplan BJ. Journal of Child and Adolescent Psychopharmacology. 2010 Apr;20(2):95-103.

FULL TEXT

Treatment of mood lability and explosive rage with minerals and vitamins: two case studies in children.

Researchers describe two medication-free boys with unstable mood and explosive rage who took a micronutrient formulation co-developed† by Hardy Nutritionals® founder David Hardy followed by reversal and re-treatment. The boys' diagnoses included pervasive developmental disorder with Asperger characteristics, attention deficit hyperactivity disorder, obsessive-compulsive disorder, and autism. Both boys benefited significantly from the micronutrient formula: mood, angry outbursts, delinquent behavior, social and attention problems, and anxious/obsessional symptoms improved when initially treated, returned when they discontinued treatment, and remitted when the micronutrient treatment was reintroduced. During a follow-up of over 2 years, both boys continued to be stable while taking the micronutrient treatment.

Treatment of mood lability and explosive rage with minerals and vitamins: two case studies in children.

Kaplan BJ, Crawford SG, Gardner B, Farrelly G. Journal of Child and Adolescent Psychopharmacology. 2002 Fall;12(3):205-19.

FULL TEXT

†The micronutrient formulation studied was a pre-2013 version of *Truehope EMPowerplus* which was co-formulated by David Hardy and Anthony Stephan. *Truehope EMPowerplus* is a registered trademark of The Synergy Group of Canada Inc., which was co-founded by David Hardy and Anthony Stephan in 1999. David Hardy officially resigned as a shareholder of The Synergy Group of Canada Inc. and director of Truehope Nutritional Support Ltd. in 2013 to focus his efforts exclusively on Hardy Nutritionals®.

STUDIES MENU

- 1-year naturalistic follow-up of a Randomised Double-Blind, Placebo-Controlled Trial ("NoMAD") Exploring the Effectiveness of Micronutrients in Improving Symptoms of Anxiety and Depression
- Gut microbiome changes with micronutrient supplementation in children with attention-deficit/hyperactivity disorder: the MADDY study
- Potential link between copy number variation and abnormal genome wide DNA methylation profile in an individual with severe ADHD and a strong response to micronutrient treatment
- Investigation of a broad-spectrum micronutrient formulation as a possible precipitant of pharmacokinetic micronutrient-drug interactions
- Broad-Spectrum Micronutrients or Antidepressants for Antenatal Depression: Effect on Maternal and Infant Birth Outcomes in an Observational Secondary Analysis of NUTRIMUM
- Efficacy and safety of a mineral and vitamin treatment on symptoms of antenatal depression: 12-week fully blinded randomised placebo-controlled trial (NUTRIMUM)
- Problems most concerning to parents of children with ADHD and emotional dysregulation in a randomized controlled trial of multinutrients: MADDY secondary analysis
- Treatment response to supplemental nutrients for ADHD is independent of diet quality: the MADDY Study RCT
- Micronutrients for ADHD in youth (MADDY) study: comparison of results from RCT and open label extension.
- Exploring the impact of antenatal micronutrients used as a treatment for maternal depression on infant temperament in the first year of life
- Household Food Insecurity Is Associated with Symptoms of Emotional Dysregulation in Children with Attention Deficit Hyperactivity Disorder: The MADDY Study.
- Effect of antenatal micronutrient or antidepressant exposure on Brazelton neonatal behavioral assessment scale (NBAS) performance within one-month of birth
- Efficacy and safety of a vitamin-mineral intervention for symptoms of anxiety and depression in adults: A randomised placebo-controlled trial "NoMAD"
- Non-pharmacological interventions for attention-deficit hyperactivity disorder in children and adolescents
- Inflammation and Vitamin C in Women with Prenatal Depression and Anxiety: Effect of Multinutrient Supplementation.
- Diet Quality as a Moderator of Response to Multinutrients for ADHD and Emotional Dysregulation: The MADDY RCT.
- Micronutrients for Attention-Deficit/Hyperactivity Disorder in Youth: A Placebo-Controlled Randomized Clinical Trial
- Paediatric adverse event rating scale: a measure of safety or efficacy? Novel analysis from the MADDY study
- Clinician guidelines for the treatment of psychiatric disorders with nutraceuticals and phytoceuticals: The World Federation of Societies of Biological Psychiatry (WFSBP) and Canadian Network for Mood and Anxiety Treatments (CANMAT) Taskforce
- Parents' priorities and preferences for treatment of children with ADHD: Qualitative inquiry in the MADDY study
- Do Changes in Blood Nutrient Levels Mediate Treatment Response in Children and Adults With ADHD Consuming a Vitamin-Mineral Supplement?
- Multinutrients for the Treatment of Psychiatric Symptoms in Clinical Samples: A Systematic Review and Meta-Analysis of Randomized Controlled Trials
- A Holistic Self-learning Approach for Young Adult Depression and Anxiety Compared to Medication-Based Treatment-As-Usual
- Resting-state networks and neurometabolites in children with ADHD after 10 weeks of treatment with micronutrients: results of a randomised placebo-controlled trial
- Novel Mineral–Vitamin Treatment for Reduction in Cigarette Smoking: A Fully Blinded Randomized Placebo-Controlled Trial
- Human gut microbiome changes during a 10 week Randomised Control Trial for micronutrient supplementation in children with attention deficit hyperactivity disorder
- Mineral-Vitamin Treatment Associated with Remission in Attention-Deficit/Hyperactivity Disorder Symptoms and Related Problems: 1-Year Naturalistic Outcomes of a 10-Week Randomized Placebo-Controlled Trial
- Multinutrients for ADHD Youth (MADDY) Trial: Preliminary Trends for Treatment
- An Observational Preliminary Study on the Safety of Long-Term Consumption of Micronutrients for the Treatment of Psychiatric Symptoms
- Can we predict treatment response in children with ADHD to a vitamin-mineral supplement? An investigation into pre-treatment nutrient serum levels, MTHFR status, clinical correlates and demographic variables
- Methylomic changes in response to micronutrient supplementation and MTHFR genotype.
- Hospitalization cost of conventional psychiatric care compared to broad-spectrum micronutrient treatment: literature review and case study of adult psychosis
- Vitamin-mineral treatment improves aggression and emotional regulation in children with ADHD: a fully blinded, randomized, placebo-controlled trial.
- Vitamin–Mineral Treatment of ADHD in Adults: A 1-Year Naturalistic Follow-Up of a Randomized Controlled Trial
- Micronutrient Treatment of Emotional Dyscontrol Following Traumatic Brain Injury
- Effect of Micronutrients on Insomnia in Adults: A Multiple-Baseline Study
- A randomised trial of nutrient supplements to minimise psychological stress after a natural disaster
- Clinically Significant Symptom Reduction in Children with Attention-Deficit/Hyperactivity Disorder Treated with Micronutrients: An Open-Label Reversal Design Study
- Vitamin–mineral treatment of attention-deficit hyperactivity disorder in adults: double-blind randomised placebo-controlled trial.
- Moderators of treatment response in adults with ADHD treated with a vitamin-mineral supplement
- Psychological functioning 1 year after a brief intervention using micronutrients to treat stress and anxiety related to the 2011 Christchurch earthquakes: a naturalistic follow-up
- Nutritional and Safety Outcomes from an Open-Label Micronutrient Intervention for Pediatric Bipolar Spectrum Disorders
- Could yeast infections impair recovery from mental illness? A case study using micronutrients and olive leaf extract for the treatment of ADHD and depression.
- Use of micronutrients attenuates cannabis and nicotine abuse as evidenced from a reversal design: a case study.
- Efficacy and cost of micronutrient treatment of childhood psychosis.
- Micronutrient formulation outperforms Berocca Performance in reducing depression, anxiety, and stress in 91 earthquake survivors.
- Feasibility of a nutritional supplement as treatment for pediatric bipolar spectrum disorders.
- Safety and tolerability of Micronutrient formulations.
- Can micronutrients improve neurocognitive functioning in adults with ADHD and severe mood dysregulation? A pilot study.
- Micronutrient formulation reduces depression, anxiety, and stress in 17 earthquake survivors with ADHD.
- Micronutrients reduce stress and anxiety following a 7.1 earthquake in adults with Attention-Deficit/Hyperactivity Disorder
- Effect of micronutrients on behaviour and mood in adults with ADHD: Evidence from an 8-week open label trial with natural extension.
- Database analysis: Micronutrient formulation greatly improves bipolar and ADHD symptoms in 161 children and adolescents.
- Successful treatment of bipolar disorder II and ADHD with a micronutrient formula: a case study.
- Micronutrient formulation outperforms medications in 88 patients with autism.
- Micronutrient formulation greatly improves brain function in rats.
- Database analysis: Micronutrient formulation greatly improves bipolar symptoms in 358 adults.
- Successful treatment of OCD with a micronutrient formula following partial response to Cognitive Behavioral Therapy (CBT): a case study.
- Multinutrient supplement as treatment: literature review and case report of a 12-year-old boy with bipolar disorder.
- Micronutrient formulation greatly improves mood, behavior, and anxiety disorders in 9 children.
- Treatment of mood lability and explosive rage with minerals and vitamins: two case studies in children.
- Micronutrient formulation greatly improves bipolar symptoms in 19 adults.
- Micronutrient formulation greatly improves bipolar symptoms in 23 adults, adolescents, and children.
- Micronutrient formulation stabilizes mood in 11 bipolar adults.

Consult with your doctor when altering your treatment regimen and before adjusting your medication dosing.

Information from Hardy Nutritionals® is for educational purposes only, and suggestions should not be considered medical advice.

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